

Pedram Hassanzadeh

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Academic Appointments

The University of Chicago

Associate (tenured) Professor (January 2024–present)

Department of the Geophysical Sciences

Committee on Computational and Applied Mathematics (CAAM)

Committee on Data Science

Data Science Institute (DSI)

PI: Climate Extremes Theory and Data (CeTD) Group

Director: AI for Climate Initiative (AICE), DSI and Institute for Climate and Sustainable Growth (ICSG)

Co-director: Human-centered Weather Forecasts Initiative (HCWF), ICSG

Rice University

Assistant and Associate (tenured) Professor (July 2016–December 2023)

Department of Mechanical Engineering and Department of Earth, Environmental & Planetary Sciences

Faculty Associate, Baker College (2017–2023)

Harvard University

Postdoctoral Fellow, Department of Earth and Planetary Sciences (2015–2016), *Mentor: Zhiming Kuang*

Ziff Environmental Fellow, Center for the Environment (2013–2015), *Hosts: Zhiming Kuang & Brian Farrell*

Woods Hole Oceanographic Institution (WHOI)

Geophysical Fluid Dynamics Program

PhD Student Fellow (summer 2012), *Supervisors: Charlie Doering & Greg Chini*

Staff Member (summers of 2014, 2017, 2018, 2022, 2023)

Education

University of California, Berkeley

PhD, Mechanical Engineering (2013)

Advisor: Phil Marcus

Thesis: *Baroclinic vortices in rotating stratified shearing flows: cyclones, anticyclones, and zombie vortices*

MA, Mathematics (2012)

Advisor: Jon Wilkening (based on work with Charlie Doering, University of Michigan)

Thesis: *Optimal transport from wall to wall*

University of Waterloo

MASc, Mechanical Engineering (2007)

Advisor: George Raithby

Thesis: *An efficient computational method for thermal radiation in participating media*

University of Tehran

BSc, Mechanical Engineering, Fluid Mechanics concentration (2005)

Honors & Awards

CAREER Award, National Science Foundation (NSF), Climate & Large-Scale Dynamics program, 2021
 Young Investigator Award, Office of Naval Research (ONR), Applied & Computational Analysis program, 2020
 Early-Career Research Fellow, The National Academy of Sciences Gulf Research Program, 2018
 Ziff Environmental Fellowship, Harvard University Center for the Environment, 2013–2015
 Postdoctoral Award for Professional Development, Office of Postdoctoral Affairs, Harvard University, 2014
 Geophysical Fluid Dynamics Fellowship, Woods Hole Oceanographic Institution, 2012
 Postgraduate Scholarship, Natural Sciences & Engineering Research Council of Canada (NSERC), 2009–2011
 Outstanding Preliminary Examination Award, ME Department, UC Berkeley, 2009
 Jonathan Laitone Memorial Scholarship, ME Department, UC Berkeley, 2009
 International Graduate Student Award, University of Waterloo, 2005–2007

Publications

Google Scholar: [link to profile](#).

* (†) indicates postdocs/grad (undergrad) students in my group. Only peer-reviewed journal papers and peer-reviewed conference papers (e.g., NeurIPS) are listed. Selected media coverage is given under each paper.

Highlighted Publications

Five key contributions across different disciplines:

- H1. Pathak J., Subramanian S., Harrington P., Raja S., Chattopadhyay* A., Mardani M., Kurth T., Hall D., Li Z., Azizzadenesheli K., [Hassanzadeh P.](#), Kashinath K. & Anandkumar A., FourCastNet: A global data-driven high-resolution weather model using adaptive Fourier neural operators, arXiv:2202.11214, 2022 ([link](#))
 Coverage: [NVIDIA Newsroom](#), [Forbes](#), [SingularityHub](#)
- H2. [Hassanzadeh P.](#), Li* W., Sun* Y.Q., Wang* J., Wikner* A., Finkel* J. & Weare J., Missing the butterfly and predicting the past: Features or bugs of accurate AI weather models?, arXiv:2608.25835, 2026, *under review* ([pre-print](#))
- H3. Sun* Y.Q., [Hassanzadeh P.](#), Zand M., Chattopadhyay A., Weare J. & Abbot D.S., Can AI weather models predict out-of-distribution gray swan tropical cyclones?, *Proceedings of the National Academy of Sciences*, 122, 2025 ([link](#))
 Coverage: [Nature \(news feature\)](#), [PNAS podcast](#), [Big Brains podcast](#), [SIAM News](#), [UChicago News](#), [Gizmodo](#), [El País](#)
- H4. Jakhar* K., Guan* Y. & [Hassanzadeh P.](#), Analytical and AI-discovered stable, accurate, and generalizable subgrid-scale closure for geophysical turbulence, *Physical Review Letters*, 136, 2026 ([link](#)) (Editors' Suggestion)
 Coverage: [APS Physics](#), [UChicago News](#), [Phys.org](#)
- H5. Subel† A., Guan* Y., Chattopadhyay* A. & [Hassanzadeh P.](#), Explaining the physics of transfer learning in data-driven turbulence modeling, *PNAS Nexus*, 2, 2023 ([link](#))
 Coverage: [IEEE Spectrum](#), [Rice News](#), [Psychology Today](#)

Peer-Reviewed Papers (Published & In Press)

70. Sun* Y.Q., [Hassanzadeh P.](#), Shaw T., Pahlavan* H.A. & Marchakitus[†] A., Predicting regional gray swans via translocation: AI weather models and Dubai's unprecedented 2024 rainfall, *Science Advances*, in press, 2026 ([pre-print](#))
Coverage: [Nature \(news feature\)](#), [SIAM News](#)
69. Lancelin* A., Wikner* A., Dubus L., Le Priol C., Abbot D.S., Bouchet F., [Hassanzadeh P.](#) & Weare J., AI-boosted rare event sampling to characterize extreme weather, *Physical Review Letters*, 137, 2026 ([link](#)) (Editors' Suggestion)
Coverage: [APS Physics](#), [Nature \(news\)](#), [Physics World](#), [UChicago News](#), [UChicago PSD](#), [Phys.org](#)
68. Jakhar* K., Guan* Y. & [Hassanzadeh P.](#), Analytical and AI-discovered stable, accurate, and generalizable subgrid-scale closure for geophysical turbulence, *Physical Review Letters*, 136, 2026 ([link](#)) (Editors' Suggestion)
Coverage: [APS Physics](#), [UChicago News](#), [Phys.org](#)
67. Guan* Y. & [Hassanzadeh P.](#), Semi-analytical eddy-viscosity and backscattering closures for 2D geophysical turbulence, *J. Fluid Mechanics*, in press, 2026 ([pre-print](#))
66. Rucker* K., Baxter I., [Hassanzadeh P.](#), Shaw T. & Pahlavan* H.A., Benchmarking regional thermodynamic trends in an AI emulator, ACE2, and a hybrid model, NeuralGCM, *Geophysical Research Letters*, 53, 2026 ([link](#))
65. Baxter I., Pahlavan* H.A., [Hassanzadeh P.](#), Rucker* K. & Shaw T., Benchmarking atmospheric circulation variability in an AI emulator, ACE2, and a hybrid model, NeuralGCM, *Geophysical Research Letters*, 53, 2026 ([link](#))
64. Guan* Y., [Hassanzadeh P.](#), Schneider T., Dunbar O., Huang D., Wu J. & Lopez-Gomez I., Online learning of eddy-viscosity and backscattering closures for geophysical turbulence using ensemble Kalman inversion, *Physical Review Research*, 8, 2026 ([link](#))
63. Guan* Y., Amoudruz L., Litvinov S., Jakhar* K., Moigani* R., Koumoutsakos P. & [Hassanzadeh P.](#), Prediction of extreme events in multiscale simulations of geophysical turbulence using reinforcement learning, *J. Computational Science*, in press, 2026 ([pre-print](#))
62. Shan W., Patel* D., Sun* Y.Q., [Hassanzadeh P.](#), Tziperman E. & Abbot D.S., Testing the limits of surface QG theory with deep learning, *J. Geophysical Research: Machine Learning and Computation*, 3, 2026 ([link](#))
61. Darman* M., [Hassanzadeh P.](#), Zanna L. & Chattopadhyay A., Fourier analysis of the physics of transfer learning for data-driven subgrid-scale models of ocean turbulence, *Machine Learning: Earth*, 2, 2026 ([link](#))
60. Nelli N., Francis D., Fonseca R., [Hassanzadeh P.](#) & Cherif C., The June 2024 Middle East compound heatwave: Dynamical drivers and AI-weather forecast models' evaluation, *Atmospheric Research*, 2026 ([link](#))
59. Bracco A., Brajard J., Dijkstra H.A., [Hassanzadeh P.](#), Lessig C. & Monteleoni C., Machine learning for the physics of climate, *Nature Reviews Physics*, 7, 2025 ([link](#))
58. Sun* Y.Q., [Hassanzadeh P.](#), Zand M., Chattopadhyay A., Weare J. & Abbot D.S., Can AI weather models predict out-of-distribution gray swan tropical cyclones?, *Proceedings of the National Academy of Sciences*, 122, 2025 ([link](#)) (featured in SIAM News)
Coverage: [Nature \(news feature\)](#), [PNAS podcast](#), [Big Brains podcast](#), [SIAM News](#), [UChicago News](#), [Gizmodo](#), [El País](#)
57. Lai C.-Y., [Hassanzadeh P.](#), Sheshadri A., Sonnewald M., Ferrari R. & Balaji V., Machine learning for climate physics and simulations, *Annual Review of Condensed Matter Physics*, 16, 2025 ([link](#))
56. Jiang R., Zhang X., Jakhar* K., Lu P.Y., [Hassanzadeh P.](#), Maire M. & Willett R., Hierarchical implicit neural emulators, *Advances in Neural Information Processing Systems (NeurIPS)*, 38, 2025 ([link](#))

55. Pahlavan* H.A., [Hassanzadeh P.](#) & Alexander M.J., On the importance of learning non-local dynamics for stable data-driven climate modeling: A 1D gravity wave-QBO testbed, *Geophysical Research Letters*, 52, 2025 ([link](#))
54. Battalio J.M., Lora J.M., Lubis* S.W. & [Hassanzadeh P.](#), Propagation and periodicity of Mars's Northern Annular Mode modulates the dust cycle, *Geophysical Research Letters*, 52, 2025 ([link](#))
53. Pahlavan* H.A., [Hassanzadeh P.](#) & Alexander M.J., Explainable offline-online training of neural networks for parameterizations: A 1D gravity wave-QBO testbed in the small-data regime, *Geophysical Research Letters*, 51, 2024 ([link](#))
52. Jakhar* K., Guan* Y., Mojjani* R., Chattopadhyay* A. & [Hassanzadeh P.](#), Learning closed-form equations for subgrid-scale closures from high-fidelity data: Promises and challenges, *J. Advances in Modeling Earth Systems*, 16, 2024 ([link](#)) (Editors' Highlight)
Coverage: [Eos Editors' Highlight](#)
51. Sun* Y.Q., Pahlavan* H.A., Chattopadhyay* A., [Hassanzadeh P.](#), Lubis* S.W., Alexander M.J., Gerber E.P., Sheshadri A. & Guan* Y., Data imbalance, uncertainty quantification, and transfer learning in data-driven parameterizations: Lessons from the emulation of gravity wave momentum transport in WACCM, *J. Advances in Modeling Earth Systems*, 16, 2024 ([link](#))
50. Mojjani* R., Chattopadhyay* A. & [Hassanzadeh P.](#), Interpretable structural model error discovery from sparse assimilation increments using spectral bias-reduced neural networks: A quasi-geostrophic turbulence test case, *J. Advances in Modeling Earth Systems*, 16, 2024 ([link](#))
49. Kruse C.G., Alexander M.J., Bramberger M., Chattopadhyay* A., [Hassanzadeh P.](#), Green B., Grimsdell A. & Hoffmann L., Recreating observed convection-generated gravity waves from weather radar observations via a neural network and a dynamical atmospheric model, *J. Advances in Modeling Earth Systems*, 16, 2024 ([link](#))
48. Maurice* M., Dasgupta R. & [Hassanzadeh P.](#), Volatile atmospheres of lava worlds, *Astronomy & Astrophysics*, 688, 2024 ([link](#))
47. Subel[†] A., Guan* Y., Chattopadhyay* A. & [Hassanzadeh P.](#), Explaining the physics of transfer learning in data-driven turbulence modeling, *PNAS Nexus*, 2, 2023 ([link](#))
Coverage: [IEEE Spectrum](#), [Rice News](#), [Psychology Today](#)
46. Chattopadhyay* A., Nabizadeh* E., Bach E. & [Hassanzadeh P.](#), Deep learning-enhanced ensemble-based data assimilation for high-dimensional nonlinear dynamical systems, *J. Computational Physics*, 477, 2023 ([link](#)) (featured in SIAM News)
Coverage: [SIAM News](#)
45. Lubis* S.W. & [Hassanzadeh P.](#), The intrinsic 150-day periodicity of the Southern Hemisphere extratropical large-scale atmospheric circulation, *AGU Advances*, 4, 2023 ([link](#)) (Editors' Highlight)
Coverage: [Eos Editors' Highlight](#), [Rice News](#), [PNNL](#)
44. Sun* Y.Q., [Hassanzadeh P.](#), Alexander M.J. & Kruse C.G., Quantifying 3D gravity wave drag in a library of tropical convection-permitting simulations for data-driven parameterizations, *J. Advances in Modeling Earth Systems*, 15, 2023 ([link](#))
43. Mojjani* R., Balajewicz M. & [Hassanzadeh P.](#), Kolmogorov n-width and Lagrangian physics-informed neural networks: A causality-conforming manifold for convection-dominated PDEs, *Computer Methods in Applied Mechanics and Engineering*, 404, 2023 ([link](#))
42. Guan* Y., Subel[†] A., Chattopadhyay* A. & [Hassanzadeh P.](#), Learning physics-constrained subgrid-scale closures in the small-data regime for stable and accurate LES, *Physica D: Nonlinear Phenomena*, 443, 2023 ([link](#))
41. Chattopadhyay* A., Pathak J., Nabizadeh* E., Bhimji W. & [Hassanzadeh P.](#), Long-term stability and generalization of observationally-constrained stochastic data-driven models for geophysical turbulence, *Environmental Data Science*, 2, 2023 ([link](#))

40. Connolly C., Barnes E.A., [Hassanzadeh P.](#) & Pritchard M., Using neural networks to learn the jet stream forced response from natural variability, *Artificial Intelligence for the Earth Systems*, 2, 2023 ([link](#))
39. Guan* Y., Chattopadhyay* A., Subel[†] A. & [Hassanzadeh P.](#), Stable a posteriori LES of 2D turbulence using convolutional neural networks: Backscattering analysis and generalization to higher Re via transfer learning, *J. Computational Physics*, 458, 2022 ([link](#))
38. Nabizadeh* E., Lubis* S.W. & [Hassanzadeh P.](#), The summertime Pacific-North American weather regimes and their predictability, *Geophysical Research Letters*, 49, 2022 ([link](#))
37. Mojgani* R., Chattopadhyay* A. & [Hassanzadeh P.](#), Discovery of interpretable structural model errors by combining Bayesian sparse regression and data assimilation: A chaotic Kuramoto-Sivashinsky test case, *Chaos*, 32, 2022 ([link](#))
36. Chattopadhyay* A., Mustafa M., [Hassanzadeh P.](#), Bach E. & Kashinath K., Towards physics-inspired data-driven weather forecasting: Integrating data assimilation with a deep spatial transformer-based U-NET in a case study with ERA5, *Geoscientific Model Development*, 15, 2022 ([link](#))
35. Khodkar* M.A. & [Hassanzadeh P.](#), A data-driven, physics-informed framework for forecasting the spatiotemporal evolution of chaotic dynamics with nonlinearities modeled as exogenous forcings, *J. Computational Physics*, 440, 2021 ([link](#))
34. Nabizadeh* E., Lubis* S.W. & [Hassanzadeh P.](#), The 3D structure of Northern Hemisphere blocking events: Climatology, role of moisture, and response to climate change, *J. Climate*, 34, 2021 ([link](#))
33. Chan P.W., [Hassanzadeh P.](#) & Kuang Z., Eddy length scale response to static stability change in an idealized dry atmosphere: A linear response function approach, *J. Atmospheric Sciences*, 78, 2021 ([link](#))
32. Lubis* S.W. & [Hassanzadeh P.](#), An eddy-zonal flow feedback model for propagating annular modes, *J. Atmospheric Sciences*, 78, 2021 ([link](#))
31. Kashinath K., Mustafa M., Albert A., Wu J.-L., Jiang C., Esmaeilzadeh S., Azizzadenesheli K., Wang R., Chattopadhyay* A., Singh A., Manepalli A., Chirila D., Yu R., Walters R., White B., Xiao H., Tchelepi H.A., Marcus P.S., Anandkumar A., [Hassanzadeh P.](#) & Prabhat, Physics-informed machine learning: Case studies for weather and climate modeling, *Philosophical Transactions of the Royal Society A*, 379, 2021 ([link](#))
30. Subel[†] A., Chattopadhyay* A., Guan* Y. & [Hassanzadeh P.](#), Data-driven subgrid-scale modeling of forced Burgers turbulence using deep learning with generalization to higher Reynolds numbers via transfer learning, *Physics of Fluids*, 33, 2021 ([link](#))
29. [Hassanzadeh P.](#), Lee C.-Y., Nabizadeh* E., Camargo S.J., Ma D. & Yeung L.Y., Effects of climate change on the movement of future landfalling Texas tropical cyclones, *Nature Communications*, 11, 2020 ([link](#)) (featured in NSF Research News)
Coverage: [NSF Research News](#), [Houston Chronicle](#), [Rice News](#), [Columbia Climate School](#)
28. Chattopadhyay* A., Subel[†] A. & [Hassanzadeh P.](#), Data-driven super-parameterization using deep learning: Experimentation with multi-scale Lorenz 96 systems and transfer learning, *J. Advances in Modeling Earth Systems*, 12, 2020 ([link](#))
27. Chattopadhyay* A., Nabizadeh* E. & [Hassanzadeh P.](#), Analog forecasting of extreme-causing weather patterns using deep learning, *J. Advances in Modeling Earth Systems*, 12, 2020 ([link](#)) (highlighted in AGU's Eos Magazine)
Coverage: [Eos](#), [Rice News](#), [NVIDIA Technical Blog](#), [The Next Web](#), [HPCwire](#)
26. Chattopadhyay* A., [Hassanzadeh P.](#) & Pasha S., Predicting clustered weather patterns: A test case for applications of convolutional neural networks to spatio-temporal climate data, *Scientific Reports*, 10, 2020 ([link](#))
25. Chattopadhyay* A., [Hassanzadeh P.](#) & Subramanian D., Data-driven prediction of a multi-scale Lorenz 96 chaotic system using machine learning methods: Reservoir computing, artificial neural network, and long short-term memory network, *Nonlinear Processes in Geophysics*, 27, 2020 ([link](#))

24. Ebad Sichani M., Anarde K.A., Capshaw K.M., Padgett J.E., Meidl R.A., Hassanzadeh P., Loch-Temzelides T.P. & Bedient P.B., Hurricane risk assessment of petroleum infrastructure in a changing climate, *Frontiers in Built Environment*, 6, 2020 ([link](#))
23. Nabizadeh* E., Hassanzadeh P., Yang D. & Barnes E.A., Size of the atmospheric blocking events: Scaling law and response to climate change, *Geophysical Research Letters*, 46, 2019 ([link](#)) (featured in NSF Research News)
Coverage: [NSF Research News](#), [Carbon Brief](#), [Rice News](#)
22. Chan P.W., Hassanzadeh P. & Kuang Z., Evaluating indices of blocking anticyclones in terms of their linear relations with surface hot extremes, *Geophysical Research Letters*, 46, 2019 ([link](#))
21. Hassanzadeh P. & Kuang Z., Quantifying the annular mode dynamics in an idealized atmosphere, *J. Atmospheric Sciences*, 76, 2019 ([link](#))
20. Khodkar* M.A., Hassanzadeh P., Nabi S. & Grover P., Reduced-order modeling of fully turbulent buoyancy-driven flows using the Green's function method, *Physical Review Fluids*, 4, 2019 ([link](#)) (Editors' Suggestion)
19. Pasha S., Hassanzadeh P., Ecker M. & Ho V., A hierarchical classification of adolescent idiopathic scoliosis: Identifying the distinguishing features in 3D spinal deformities, *PLoS One*, 14, 2019 ([link](#))
18. Khodkar* M.A. & Hassanzadeh P., Data-driven reduced modelling of turbulent Rayleigh-Bénard convection using DMD-enhanced Fluctuation-Dissipation Theorem, *J. Fluid Mechanics*, 852, 2018 ([link](#))
17. Ronalds B., Barnes E.A. & Hassanzadeh P., A barotropic mechanism for the response of jet stream variability to Arctic Amplification and sea ice loss, *J. Climate*, 31, 2018 ([link](#))
16. Mahdinia M., Hassanzadeh P., Marcus P.S. & Jiang C.-H., Stability of 3D Gaussian vortices in an unbounded, rotating, vertically-stratified Boussinesq flow: Linear analysis, *J. Fluid Mechanics*, 824, 2017 ([link](#))
15. Jeevanjee N., Hassanzadeh P., Hill S. & Sheshadri A., A perspective on climate model hierarchies, *J. Advances in Modeling Earth Systems*, 9, 2017 ([link](#))
14. Ma D., Hassanzadeh P. & Kuang Z., Quantifying the eddy-jet feedback strength of the annular mode in an idealized GCM and reanalysis data, *J. Atmospheric Sciences*, 74, 2017 ([link](#))
13. Anderson B.T., Hassanzadeh P. & Caballero R., Persistent anomalies of the extratropical Northern Hemisphere wintertime circulation as an initiator of El Niño/Southern Oscillation events, *Scientific Reports*, 7, 2017 ([link](#))
12. Hassanzadeh P. & Kuang Z., The linear response function of an idealized atmosphere. Part II: Implications for the practical use of the Fluctuation-Dissipation Theorem and the role of operator's nonnormality, *J. Atmospheric Sciences*, 73, 2016 ([link](#))
11. Hassanzadeh P. & Kuang Z., The linear response function of an idealized atmosphere. Part I: Construction using Green's functions and applications, *J. Atmospheric Sciences*, 73, 2016 ([link](#))
10. Hassanzadeh P. & Kuang Z., Blocking variability: Arctic Amplification versus Arctic Oscillation, *Geophysical Research Letters*, 42, 2015 ([link](#))
9. Marcus P.S., Pei S., Jiang C.-H., Barranco J.A., Hassanzadeh P. & Lecoanet D., Zombie vortex instability. I. A purely hydrodynamic instability to resurrect the dead zones of protoplanetary disks, *Astrophysical J.*, 808, 2015 ([link](#))
8. Hassanzadeh P., Chini G.P. & Doering C.R., Wall to wall optimal transport, *J. Fluid Mechanics*, 751, 2014 ([link](#))
7. Hassanzadeh P., Kuang Z. & Farrell B.F., Responses of midlatitude blocks and wave amplitude to changes in the meridional temperature gradient in an idealized dry GCM, *Geophysical Research Letters*, 41, 2014 ([link](#)) (AGU Research Spotlight & Editors' Highlight)
Coverage: [Eos Research Spotlight](#)

6. Marcus P.S., Pei S., Jiang C.-H. & Hassanzadeh P., Three-dimensional vortices generated by self-replication in stably stratified rotating shear flows, *Physical Review Letters*, 111, 2013 ([link](#))
Coverage: [Berkeley News](#)
5. Hassanzadeh P., Marcus P.S. & Le Gal P., The universal aspect ratio of vortices in rotating stratified flows: theory and simulation, *J. Fluid Mechanics*, 706, 2012 ([link](#))
4. Hassanzadeh P. & Raithby G.D., Efficient iterative solution of the P1 equation, *J. Heat Transfer*, 131, 2009 ([link](#))
3. Hassanzadeh P., Raithby G.D. & Chui E.H., Efficient calculation of radiation heat transfer in anisotropically scattering media using the QL method, *J. Computational Thermal Sciences*, 1, 2009 ([link](#))
2. Hassanzadeh P. & Raithby G.D., Application of the finite volume method to the second-order radiative transfer equation: accuracy and solution cost, *J. Numerical Heat Transfer-B*, 53, 2008 ([link](#))
1. Hassanzadeh P., Raithby G.D. & Chui E.H., The efficient calculation of radiation heat transfer in participating media, *J. Thermophysics and Heat Transfer*, 22, 2008 ([link](#))

Under Review

- Hassanzadeh P., Li* W., Sun* Y.Q., Wang* J., Wikner* A., Finkel* J. & Weare J., Missing the butterfly and predicting the past: Features or bugs of accurate AI weather models?, *under review* ([pre-print](#))
- Ainslie C., Hassanzadeh P., Mahoney M.W. & Chattopadhyay A., Eigenanalysis framework for autoregressive neural emulators of multi-scale chaotic dynamics, *under review* ([pre-print](#))
- Asch* A., Rossellini R., Hassanzadeh P. & Willett R., Rigorous uncertainty quantification of probabilistic AI weather forecasts with conformal prediction, *under review* ([pre-print](#))
- Keutgen De Greef M., Resplandy L., Champenois B., Poupon M., Li* W., Hassanzadeh P. & Zanna L., SamudraBGC: Machine learning emulation of regional mesoscale ocean biogeochemistry, *under review* ([pre-print](#))
- Aitken C., Masiwal* R., Marchakitus[†] A., Kowal* K., Gupta M., Yang T., Jina A., Hassanzadeh P., Boos W.R. & Kremer M., Designing probabilistic AI monsoon forecasts to inform agricultural decision-making, *under review* ([pre-print](#))
- Masiwal* R., Aitken C., Marchakitus[†] A., Gupta M., Kowal* K., Pahlavan* H.A., Yang T., Sun* Y.Q., Kremer M., Jina A., Boos W.R. & Hassanzadeh P., Decision-oriented benchmarking to transform AI weather forecast access: Application to the Indian monsoon, *under review* ([pre-print](#))
- Zhou A., Wikner* A., Lancelin* A., Hassanzadeh P. & Farimani A.B., Reframing generative models for physical systems using stochastic interpolants, *under review* ([pre-print](#))
- Yao L., Yang D., Duncan J.P.C., Chattopadhyay A., Hassanzadeh P., Bhimji W. & Yu B., Deep learning the sources of MJO predictability: A spectral view of learned features, *under review* ([pre-print](#))
- Chattopadhyay* A., Sun* Y.Q. & Hassanzadeh P., Challenges of learning multi-scale dynamics with AI weather models: Implications for stability and one solution, *under review* ([pre-print](#))

Oral Presentations

2026: University of Oxford, Atmospheric, Oceanic and Planetary Physics seminar (invited); Max Planck Institute for Biogeochemistry, Jena (invited); Colloquium for the editors of the American Physical Society journals (invited; virtual); APS Global Physics Summit (March Meeting), Topical Group on Physics of Climate tutorial (invited tutorial); Discussion Meeting on Future Connections between Turbulence, Quantum Computing and Generative Machine Learning, Weimar (invited); University of Reading, VerAI Workshop on Verification of AI Weather Models (invited); SIAM Conference on Uncertainty Quantification (invited minisymposium talk); Gates

Foundation Nimbus Program Convening (invited); Rossbypalooza Summer School, UChicago (invited lecture); SciFM26 Foundation Models and AI Agents for Science Conference, UChicago (panelist); SIAM Conference on Mathematics of Planet Earth; AMS Annual Meeting

2025: University of Cambridge, DAMTP seminar (invited); ETH Zürich, Institute for Atmospheric and Climate Science seminar (invited); Max Planck Institute for Meteorology, CELLO International Conference on Earth System Modeling (invited); Caltech-UChicago AI+Science Conference (invited); UK Met Office AI Seminar Series (invited; virtual); ECMWF Using ECMWF's Forecasts (UEF) meeting (invited); Brown University, SMARTIES seminar (invited; virtual); University of Illinois Urbana-Champaign, Climate, Meteorology & Atmospheric Sciences seminar (invited); University of Maryland, ESSIC seminar (invited; virtual); University of Maryland, Brin Mathematics Research Center workshop on Contemporary Issues and Future Opportunities in Applied Mathematics (invited); University of Lausanne, Expertise Center for Climate Extremes seminar (invited); IIT Bombay, Workshop on Data-Driven Methods for Weather, Climate, and Sustainability (invited); University of Leicester, workshop on Statistical Mechanics of the Climate System and of Ecosystems (invited lecture); SIAM Conference on Mathematical and Computational Issues in the Geosciences (invited minisymposium talk; virtual); AGU Fall Meeting (2 invited talks); AIM for Scale/Gates Foundation Weather Partnership Convening, Nairobi (invited); WMO Conference on AI for Weather Prediction, Abu Dhabi (panelist); SIAM Annual Meeting (minisymposium talk)

2024: Northwestern University, Engineering Sciences & Applied Mathematics seminar (invited); University of Notre Dame, Environmental Fluid Dynamics seminar (invited); University of Wisconsin-Madison, workshop on Advancing Fluid and Soft-Matter Dynamics with Machine Learning and Data Science (invited); Portland State University, Mathematics Colloquium and RTG seminar (2 talks; invited); SIAM Mathematics of Planet Earth Activity Group webinar (invited; virtual); University of Leeds, EPSRC NFFDy Summer Programme on Data-Driven Fluid Dynamics (3 invited lectures); University of Michigan, SciFM Summer School (invited lecture; virtual); UChicago AI+Science Summer School (invited lecture); Rossbypalooza Summer School, UChicago (invited lecture); UChicago, EPIC Lunch Series (invited); UChicago Research Computing Center, Mind Bytes symposium (invited); ICCS Cross-VESRI Meeting, University of Cambridge; US CLIVAR Atmospheric Blocking Workshop; SIAM Conference on Mathematics of Planet Earth (minisymposium talk); SIAM Conference on Uncertainty Quantification; AGU Fall Meeting; AMS Conference on Atmospheric & Oceanic Fluid Dynamics; APS March Meeting

2023: NYU Courant Institute, Center for Atmosphere Ocean Science colloquium (invited); NYU Courant Institute, Center for Data Science (invited); ENS Paris (invited); Georgia Tech, Earth & Atmospheric Sciences seminar (invited); UC Santa Barbara, Applied Mathematics seminar (invited); UChicago, Committee on Computational & Applied Mathematics colloquium (invited); UChicago, Department of the Geophysical Sciences seminar (invited); Google (invited); Southern Methodist University, Data Science Institute seminar (invited); University of New Hampshire (invited); PhilFest symposium in honor of Philip Marcus, UC Berkeley (invited); APS Topical Group on Physics of Climate seminar series (invited; virtual); ITU AI for Good series on AI and Climate Science (invited; virtual); WHOI Geophysical Fluid Dynamics Program (lecture); International Congress on Industrial & Applied Mathematics; SIAM Conference on Computational Science and Engineering; AGU Fall Meeting; AMS Annual Meeting; APS March Meeting; APS Division of Fluid Dynamics Meeting

2022: MIT, Program in Atmospheres, Oceans and Climate seminar (invited); Stanford University, Climate Dynamics seminar (invited); Columbia University, LEAP Center (invited); UC San Diego, Scripps Institution of Oceanography (invited); Johns Hopkins University, Center for Environmental & Applied Fluid Mechanics seminar (invited); Mathematisches Forschungsinstitut Oberwolfach workshop (invited); Penn State University, Applied Mathematics seminar (invited); UC Santa Cruz, Applied Mathematics seminar (invited); Carnegie Mellon University, AI for Super-Resolution Simulations Workshop (invited); US Naval Academy, Mathematics Department (invited); IMSI Workshop on Machine Learning for Climate and Weather Applications, UChicago (invited); AMS (American Mathematical Society) Fall Western Sectional Meeting (invited); SIAM Conference on Mathematics of Planet Earth (invited); San Francisco State University, Physics & Astronomy colloquium (invited; virtual); Rice SPEED Center conference (invited); WHOI Geophysical Fluid Dynamics Program (lecture); WCRP Model Hierarchies Workshop, Stanford University; SIAM Conference on Mathematics of Data Science (2 minisymposium talks); SIAM Texas-Louisiana Section Meeting; AGU Fall Meeting (2 talks); Ocean Sciences

Meeting (virtual); AMS Annual Meeting (virtual); AMS Conference on Atmospheric & Oceanic Fluid Dynamics; APS March Meeting; APS Division of Fluid Dynamics Meeting

2021: Caltech, CliMA seminar (invited); Caltech, CliMA seminar (invited; virtual); Yale University, Atmosphere, Ocean & Climate Dynamics seminar (invited; virtual); KITP Program on Machine Learning for Climate Physics, UC Santa Barbara (invited); NVIDIA (invited; virtual); Goethe University Frankfurt, Theory of Atmospheric Dynamics & Climate seminar (invited; virtual); AGU Fall Meeting (invited; virtual); SIAM Annual Meeting (invited; virtual); US National Congress on Computational Mechanics (invited; virtual); Workshop on Knowledge Guided Machine Learning (invited; virtual); IS-ENES₃/ESiWACE₂ Machine Learning Workshop (invited; virtual); Charlie Doering GFD Appreciation Symposium (invited; virtual); SIAM Conference on Applications of Dynamical Systems (virtual); Atmospheric Blocking Workshop (virtual); AMS Annual Meeting (virtual); APS Division of Fluid Dynamics Meeting

2020: Stanford University, Climate Dynamics seminar (invited; virtual); Columbia University, Machine Learning in Science & Engineering Conference (invited; virtual); NYU Courant Institute, Atmosphere Ocean Science colloquium (invited; virtual); University of Washington, Atmospheric Sciences colloquium (invited; virtual); Purdue University, Earth, Atmospheric & Planetary Sciences seminar (invited); UC Irvine, Earth System Science seminar (invited; virtual); UC Davis, Atmospheric Science seminar (invited; virtual); NRL/ONR Data Assimilation Workshop (invited; virtual); Mitsubishi Electric Research Laboratories (invited); University of Houston, Civil & Environmental Engineering seminar (invited); University of Texas at El Paso, Mechanical Engineering seminar (invited; virtual); Rice University, Scientia Institute (invited; virtual); Rice Ken Kennedy Institute HPC Conference (invited); AGU Fall Meeting (invited; virtual); SIAM Conference on Mathematics of Planet Earth (invited; virtual); AGU Fall Meeting (contributed; virtual); AMS Annual Meeting (2 talks); EGU General Assembly (virtual); APS Division of Fluid Dynamics Meeting (virtual)

2010–2019: Harvard University Center for the Environment (invited); MIT, MASS seminar (invited); Stanford University, Climate Dynamics seminar (invited); Caltech (invited); Columbia University, SEAS Colloquium in Climate Science (invited); UC Berkeley, Earth & Planetary Science colloquium (invited); UC Berkeley, Fluids seminar (invited); UCLA, Atmospheric & Oceanic Sciences seminar (invited); University of Oxford, Palmer group seminar (invited); Colorado State University, Atmospheric Science colloquium (invited); Texas A&M University, Atmospheric Sciences seminar (invited); IUTAM Workshop on Stochastic Approaches to Transitions in Fluid Flows, Cornell University (invited); International Congress on Industrial & Applied Mathematics (invited); SIAM Conference on Mathematics of Planet Earth (invited); SIAM Conference on Applications of Dynamical Systems (invited); SIAM Conference on Uncertainty Quantification (invited); US Naval Academy, Mathematics Department (2 talks; invited); University of Houston, Mathematics Department (invited); University of Houston, Mechanical Engineering seminar (invited); Rice University, Computational & Applied Mathematics seminar (invited); Rice University, Earth Science seminar (invited); Rice University, Mechanical Engineering seminar (invited); Rice Ken Kennedy Institute (invited); Rice SSPEED Center conference (invited); Rice SSPEED Center Harvey Conference (invited); University of Oxford, Machine Learning for Weather and Climate Workshop; University of Reading, Workshop on Atmospheric Blocking; UCLA, Workshop on Connecting Theory to Experiments in Geophysical & Astrophysical Fluid Dynamics; NCAR IMAGE Theme-of-the-Year Turbulence workshop; WHOI Geophysical Fluid Dynamics Program; Texas Systems Day, Texas A&M University; AGU Fall Meeting; AMS Annual Meeting; AMS Conference on Atmospheric & Oceanic Fluid Dynamics; APS March Meeting; APS Division of Fluid Dynamics Meeting

Advising & Mentoring

Postdoctoral Fellows and Research Scientists

Current: Dr. Katie Kowal (Research Scientist and Director of AI Weather), Dr. Bing Gong (Research Scientist), Dr. Alex Wikner (Schmidt AI in Science Postdoctoral Fellow), Dr. Justin Finkel (AICE-DSI Postdoctoral Fellow), Dr. Jess Wan (CSEi Postdoctoral Fellow), Dr. Jhayron S. Pérez-Carrasquilla (AICE-DSI Postdoctoral Scholar), Dr. Rajat Masiwal, Dr. Lualawi Mareshet Admasu

Dr. Y. Qiang Sun (2022–2026): Associate Professor of Atmospheric Sciences, Nanjing University

Dr. Yifei Guan (2020–2024): Assistant Professor of Mechanical Engineering, Union College

Dr. Lei Wang (2017–2018; joint with Zhiming Kuang, Harvard): Assistant Professor of Earth, Environmental & Planetary Sciences, Purdue University

Dr. Dhruvit Patel (NSF Postdoctoral Fellow, 2024–2025): Founding Research Scientist at Silurian AI

Dr. Rambod Mojangani (2021–2024): ML Research Engineer at RTX

Dr. Cedric Gillmann (2021–2023): Senior Researcher at ETH Zürich

Dr. Maxime Maurice (2021–2023): Scientist at LMD, IPSL

Dr. Sandro Lubis (2019–2022): Research Scientist at Pacific Northwest National Laboratory

Dr. Amin Khodkar (2017–2019): Postdoc at UCLA

PhD Students

Current: Anna Asch (CAAM), Tina Wang (CAAM), Katy Rucker (co-advised with Tiffany Shaw), Weidong Li, Zhixing Liu (co-advised with Dorian Abbot)

Dr. Ashesh Chattopadhyay (2017–2022): Assistant Professor of Applied Mathematics, UC Santa Cruz

Dr. Karan Jakhar (2021–2025): AI+Weather Research Scientist at Pravah

Dr. Ebrahim Nabizadeh (2017–2022): ML Scientist at Verisk

Master's and Undergraduate Students

Current: Daniel Boscu

Lichuan Xu (MS, 2024–2026)

Willow Stenglein (summer 2025): undergraduate at UT Austin

Adam Marchakitus (undergraduate researcher, 2024–2025): Research Staff, UChicago AI for Climate Initiative

Moein Darman (MS, 2022–2024): PhD student at UC Santa Cruz

Adam Subel (undergraduate researcher, 2018–2021; NSF GRFP recipient and Rice Distinguished Research Excellence Award): PhD student at NYU Courant Institute

David Lee (MS, 2018–2020; supported by US Air Force): Pilot, US Air Force

Joey Lou (undergraduate researcher, 2018–2019): Graduate student at UC Berkeley

Sebastian Jia (undergraduate researcher, summer 2017): Graduate student at the University of Pennsylvania

Service

Editorial Boards

Associate Editor, Science Advances (AAAS), 2026–present

Associate Editor, Journal of Advances in Modeling Earth Systems (AGU), 2022–2026

Co-editor, Weather and Climate Dynamics (EGU), 2020–2026

Editorial Board, Environmental Research: Climate (IOP)

Professional Societies

Chair, Society for Industrial and Applied Mathematics (SIAM) Activity Group on Mathematics of Planet Earth (MPE), 2025–present

Member-at-Large, American Physical Society (APS) Topical Group on the Physics of Climate (GPC), 2025–2026

Chair, APS-GPC Seminar Series Committee, 2021–2023

Member, Nominations Committee, APS-GPC, 2025

Member, Honors Selection Committee, APS-GPC, 2022–2023

Advisory Boards & Panels

Member, Joint Advisory Group on Artificial Intelligence (JAG-AI), World Meteorological Organization (WMO), 2025–present

Member, Steering Committee, UCAR/CPAESS Climate and Global Change Postdoctoral Fellowship Program, 2026–2029

Member, Weather Forecasts for Farmers Technical Panel, Agriculture Innovation Mechanism (AIM) for Scale (a Gates Foundation-UAE initiative), 2024–present

Member, NVIDIA Advisory Council on Physics-ML for Climate Science, 2024–2026

Conference, Workshop & Session Organization

Member, Organizing Committee, *UChicago-Caltech AI+Science Conference*, Chicago (2026)

Co-organizer, IMSI workshop *Unpacking AI Weather Emulators: What and How Do They Learn?*, UChicago (2026)

Co-organizer, *UChicago Workshop on AI Weather Forecasting for Developing Economies*, Data Science Institute (2026)

Organizer, *AI for Climate Initiative Speaker Series*, UChicago (2025–present)

Co-organizer, *Statistical Mechanics of the Climate System and of Ecosystems* workshop (StatPhys29 satellite), University of Leicester (2025)

Co-organizer, *AI Weather Forecasting Training Program* for meteorological and agriculture agencies of low- and middle-income countries, HCWF (2025–present)

Organizer, APS-GPC monthly *Virtual Seminar Series on Climate Physics* (2022–2023)

Mini-symposium organizer: *SIAM Mathematics of Data Science Conference* (2022); *SIAM Mathematics of Planet Earth Conference* (2022)

Session organizer: *AGU Fall Meeting* (2017–2021); *AMS Annual Meeting* (2022, 2 sessions); *US National Congress on Computational Mechanics* (2021); *AGU Fall Meeting* (2018); *APS-DFD Annual Meeting* focus session (2018)

Session chair: *WCRP Model Hierarchies Workshop*, Stanford (2022); *AI Super-Resolution Simulations Workshop*, CMU (2022); *KITP Machine Learning and the Physics of Climate* workshop (2021); *APS-DFD Annual Meeting* (2013, 2018, 2022); *Modeling Hierarchies Workshop*, Princeton (2016); *AMS-AOFD Meeting* (2015)

University Service

The University of Chicago (2024–present)

Member, Executive Steering Committee, joint ICSG-DSI Program on Climate and AI, 2026–present
 Member, Faculty Advisory Board for Computing and IT, Physical Sciences Division, 2026–present
 Core Faculty, Climate Systems Engineering initiative (CSEi), 2025–present
 Chair, Postdoc Advising Committee, Department of the Geophysical Sciences, 2026–present
 Member, Curriculum Committee, Department of the Geophysical Sciences, 2026–present
 Member, Colloquia Committee, Department of the Geophysical Sciences, 2025–2026
 Chair, Graduate Application Screening Committee, Department of the Geophysical Sciences, 2025–2026
 Member, UChicago Working Group for Sub-Saharan Africa, 2025–present
 Member, Postdoctoral Scholars Search Committee, DSI, 2024

Rice University (2016–2023)

Steering committee, Gulf Scholars Program (2021–2023)
 Faculty search committees: Earth, Environmental & Planetary Sciences (2021–2022); Civil & Environmental Engineering (2018, 2019)
 School of Engineering Strategic Planning committee (2017–2018); Marshals committee (2017–2019, 2021)
 Mechanical Engineering: Chair, Seminar Series committee (2021–2023); Awards committee (2021–2023); Co-chair, Seminar Series committee (2018–2020); Graduate Studies committee (2016–2021)
 Keynote, Gulf Coast Undergraduate Research Symposium (2016); Judge, Rice Undergraduate Research Symposium (2016–2018, 2020)

Teaching

GEOS 35450/25450: Foundations of AI for Geosciences (Fall 2026)
 GEOS 35500/25500 and DS 35510: Scientific Machine Learning for Climate and Chaos (Spring 2025, 2026)
 MECH 575: Introduction to Hydrodynamic Stability (Spring 2017, 2018, 2019, 2020, 2022)
 MECH 611/612: Independent Study on Geophysical Fluid Dynamics (Spring & Fall 2018)
 GEOS 13300: Atmosphere (Fall 2025)
 MECH 371: Fluid Mechanics (Fall 2017, 2018, 2019, 2020, 2021)

Summer Schools & Short Courses

UChicago AI+Science Summer School (2024)
 University of Leeds EPSRC NFFDy Summer Programme (2024)

Selected Media Coverage & Interviews

AI and the physics of weather and climate: Nature (news feature, 2025); Big Brains podcast (UChicago, 2026); PNAS podcast (2025); FOX Weather (TV, 2026); SIAM News (invited article, 2025); CBS News Chicago (TV, 2026); MaRS Magazine (2026); FOX 32 Chicago (TV, 2026); Communications of the ACM (2026); The Chicago Maroon (2025); El País (2025)

Human-centered weather forecasting: Nature Climate Change (2025); The Economist (2025); Wall Street Journal (2025); Le Figaro (2025); The Hindu BusinessLine (2025); Berkeley News (2025); SciDev.Net (2025)

Profiles and other: Faculty Focus, UChicago Physical Sciences Division (2024); Rice News: NSF CAREER Award (2021); Sustainability by Numbers, Hannah Ritchie (2025)